PABLO ERNESTO GÓMEZ IGNOROSA

Mexico City, Mexico

pgomezignorosa@gmail.com · *LinkedIn* · *GitHub*

EDUCATION

Instituto Tecnológico Autónomo de México (ITAM) — Mexico City, Mexico

Ingeniería en Computación (Computer Engineering) | 2023 – Expected Dec. 2027

GPA: 9.0/10 overall · 9.6/10 major coursework · 80% Merit Scholarship (maintained throughout award period)

Relevant Coursework: Machine Learning, Data Mining, Probability, Linear Algebra, Simulation & Monte Carlo Methods, Numerical Methods, and Computer Organization.

ACADEMIC EXPERIENCE

Facultad Menor de Computación y Ciencia de Datos — ITAM

Peer Tutor & Faculty Collaborator | Spring 2026 – Present · 10 hrs/week

- Provide undergraduate peer tutoring in data management, algorithmic reasoning, and advanced data structures.
- Collaborated with Prof. Marco Antonio Morales Aguirre on the redesign and deployment of ITAM Robotics' website using React/CSS and Vercel.
- Currently collaborate with Prof. Alberto Elías Petrilli Barceló, developing foundational experience with ROS 2 and computer vision for robotics.

LEADERSHIP & SERVICE

Makers — ITAM

Software Development Team Lead / Volunteer | 2024 – Present

- Progressed from junior member to Software Development Team Lead for two semesters; continue as a volunteer after completing university service requirements.
- Led and contributed to interactive campus maps, internal scheduling systems, and inventory tools supporting the organization's 3D-printing services.

SELECTED PROJECTS

Lyceum — Educational Knowledge & Document Analysis Platform

Python · Django · React · PostgreSQL · Neo4j | *Demo*

- Developing a research-oriented platform to analyze academic documents and represent concept relationships through graph-based structures; current work focuses on document segmentation, clustering, and graph-based knowledge representation.

Finitio — Interactive University Mathematics Platform

Django · React · PostgreSQL · Interactive Visualizations | *Demo*

- Co-created a university mathematics learning platform with interactive visualizations; built courses on Bayes' Theorem and Linear Transformations and a reusable content system for courses and mathematical graphics.

NASA Turbofan Predictive Maintenance — Data Mining Project

Python · scikit-learn · XGBoost · Pandas · NumPy | *GitHub*

- Built a predictive-maintenance pipeline using NASA C-MAPSS simulated sensor data, including RUL/failure feature engineering and comparison of Logistic Regression, Random Forest, and XGBoost.
- Optimized XGBoost through cross-validation and cost-sensitive thresholding, reaching 95.8% recall for imminent failures in the simulated test set.

TECHNICAL SKILLS

Programming: Python, C++, Java, R, MATLAB, SQL · Frameworks/ML: Django, React, Angular, scikit-learn, XGBoost

Data & Tools: PostgreSQL, SQLite, Neo4j, Git/GitHub, Jupyter, Vercel

LANGUAGES

Spanish: Native · English: Advanced — Cambridge English C1